With special thanks to the sponsorship of

Tour de Maths 2022

Leg 4

DAINFERN COLLEGE

Question Paper

With special thanks to the sponsorship of

5 Mark Questions

Question 1

A rectangle is divided into four rectangles, with the following areas:

What is the area of rectangle A?

Question 2

In a jazz band, Bob plays the trumpet, John plays the saxophone and Susan sings. They are all the same age. There are 3 more members of the jazz band, who are 19, 20 and 21 years old, respectively.

The average age of the band is 21. How old is Susan?

Question 3

Which of the following fractions is the average of the other four?

(a) $\frac{1}{2}$ (b) $\frac{5}{9}$ (c) $\frac{7}{18}$ (d) $\frac{4}{9}$ (e) $\frac{8}{9}$

Question 4

Jack and Jill wrote a test. Jack answered each question in 3 minutes whilst Jill answered each question in 1 minute. Jill took a nap for an hour in the middle of the test and they still finished at the same time. How many questions were in the test?

Question 5

At a particular bank, you can trade dollars, pounds and rands. The bank offers the following rates:

- a dollars for b pounds
- c dollars for d rands

How many rands will you get for e pounds?

(a) $\frac{bcd}{ae}$ (b) $\frac{bde}{ac}$ (c) $\frac{ace}{bd}$ (d) $\frac{bec}{ad}$ (e) $\frac{ade}{bc}$

Question 6

When a barrel is 70% full it contains 30 litres more water than when it is 30% full. How many litres does the barrel hold when it is full?

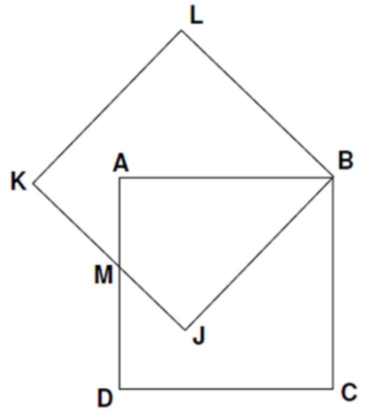
Question 7

Find the 111th term of the sequence: $1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5, 6, 6, ...$

Question 8

In the following image, $ABCD$ and $BJKL$ are two identical squares of side length $6cm$.

M is the midpoint of AD and JK. What is the area of the hexagon $BCDMKL$ $(in\ cm^2)$?



Question 9

Given that $a + b = 12$ and $3ab = 6$, find the value of $a^2 + b^2$.

Question 10

$$\boxed{\sqrt{PI} + E = \sqrt{PIE}}$$

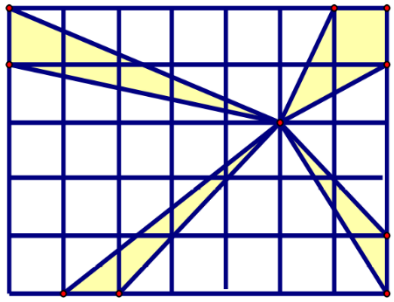
P, I and E represent different digits from $0 - 9$, and $P \neq 0$. Find the values of P, I and E and write down the equation.

10 Mark Questions

Question 11

30 squares of area 1 square unit are put together to create a rectangle as shown.

What is the area of the shaded region in square units?



Question 12

What is the last digit of $3^{2022} + 2^{2022} + 5^{2022}$?

Question 13

Let $\lfloor x \rfloor$ denote the greatest integer less than or equal to x. Find the value of:

$$\left\lfloor \frac{1}{10} \right\rfloor + \left\lfloor \frac{2}{10} \right\rfloor + \left\lfloor \frac{3}{10} \right\rfloor + \cdots + \left\lfloor \frac{99}{10} \right\rfloor$$

Question 14

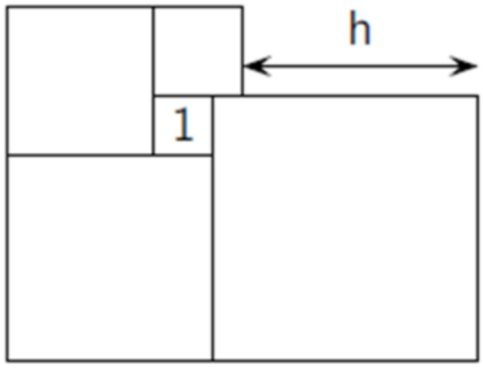
At the begging of a party, there are n people at the party. A bit later, 31 women leave the party, leaving twice as many men as women. Then, 55 men leave, leaving three times as many women as men in the party. What is the number n?

Question 15

Five squares are placed together as shown alongside.

The small square indicated has an area of 1.

What is the value of h?



20 Mark Questions

Question 16

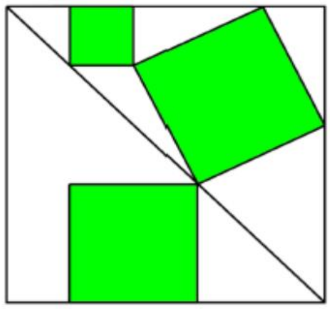
In a town, there are 21 heroes who always tell the truth and 2000 thieves who always lie. The town leader divided 2020 of these 2021 people into 1010 pairs. Every person in the pair described the other as either a hero or a thief.

In the end, 2000 people were called heroes and 20 were called thieves.

How many pairs of two thieves were there?

Question 17

The outer shape drawn is a square. Three squares are drawn inside as shown. Determine the ratio of the sum of the areas of the shaded squares to the area of the largest square.



Question 18

An octagonal swimming pool is built. Its side lengths are consecutively 10m, 20m, 30m, 40m, 50m, 60m, 70m and 80m.

All the pool's angles are right angles. Find the top surface area of the pool in square metres.

Question 19

Five numbers are added two at a time in all possible ways. The ten sums are:

7, 12, 13, 14, 15, 20, 23, 24, 29, 31.

Find the five numbers.

Question 20

Solve for x if $\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x+\cdots}}}} = \frac{5}{2}$

TOTAL = 200